

Quantum Codes for Private Distributed Matrix Multiplication in the High Privacy Regime

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Abstract—We consider the problem of quantum private distributed matrix multiplication (QPDMM) for outer product partitioning (OPP). In this setting, a user has two large matrices, which are divided into K and L sub-matrices, respectively, and has no or limited computational resources. Thus, the user relies on N servers to do the required computation. Any $T < N$ servers can be colluding to decode the matrices that the user prefers to keep private. The goal for PDMM and QPDMM is to find the least number of servers that can do the required operation, i.e., confine the interference subspace to the lowest dimension possible. It was shown previously that, in order to extend the classical codes for PDMM to their quantum counterparts, a feasibility condition is required. The feasibility condition is not always satisfied, and thus, new quantum codes for PDMM need to be designed. In this paper, we focus on one of the cases where the feasibility condition is not satisfied, namely, the high privacy regime. The high privacy regime is the regime that focuses on the following two cases, $T \geq K \geq L$ and $K \geq T \geq L$.

I. INTRODUCTION

Artificial intelligence and machine learning have been rapidly growing for the past decade. Prior to this growth, edge devices could perform certain tasks without requiring high computational power. As a result of the aforementioned growth, computationally intensive tasks such as matrix multiplication (MM) are required to train models using high computational power, which is not available on edge devices. This is mainly due to the high dimensionality of the feature vectors and model parameters. At the same time, the matrices involved in the MM task often contain sensitive and personal information that the user does not like to share, making privacy a fundamental requirement.

Private distributed matrix multiplication (PDMM) was introduced to address this problem by enabling a user to securely (privately) outsource MM tasks while ensuring information-theoretic privacy against curious servers [1]. Since its initial formulation, PDMM has evolved into a rich research area, with numerous coding techniques, privacy models, and system formulations proposed in the literature; see, e.g., [2]–[13].

As a general framework for the PDMM setting, the user sends the two matrices to N servers for the computation task. Since any T servers can exchange the user's data, i.e., collude, the information sent should be private from any T colluding servers. The honest but curious servers respond back over a classical channel to the user in a way that the user can decode the required multiplication. The main difference between the

classical and quantum PDMM settings is the presence of an entangled state shared among the servers and the presence of quantum channels for the responses of the servers.

In this work, we focus on the outer product partitioning (OPP) variant of PDMM. In the OPP setting, the matrix A is partitioned *horizontally* into K sub-blocks, while the matrix B is partitioned *vertically* into L sub-blocks. The objective is to compute the product AB privately in the presence of any T colluding servers. Among the existing state-of-the-art schemes for this setting, three polynomial-code-based constructions have emerged as the leading approaches: gap additive secure polynomial (GASP), cyclic-addition degree tables (CAT), and discretely optimized GASP (DOG) [14]–[16]. These schemes represent the current benchmark for rate performance in classical OPP-based PDMM.

In a recent work, it was shown that these classical polynomial codes can indeed be extended to their quantum counterpart if a certain feasibility condition is satisfied [17]. The feasibility of this extension depends explicitly on the system parameters K , L , and T . In the case that the feasibility condition is satisfied, it was shown that $T = O(KL)$. In this paper, we focus on the complementary and previously unresolved regime in which this feasibility condition is *not* satisfied, i.e., $T = o(KL)$. Our goal is to design quantum coding schemes that overcome the limitations of direct extensions of the schemes developed for the classical case, and fully exploit the available quantum resources.

The relationship among the system parameters naturally divides the problem into two main regimes (without loss of generality assume $K \geq L$ [15]): the *high privacy regime*, where $T \geq K \geq L$ or $K \geq T \geq L$, and the *low privacy regime*, where $K \geq L \geq T$. Each regime has its own approach for designing a suitable quantum code. In this paper, we focus on the high privacy regime. We construct polynomial quantum codes that minimize the number of required servers, thus maximizing the achievable rate, which is defined formally in the subsequent section for the high privacy regime. The approach for designing quantum codes for the low privacy regime is technically different, thus we refrain from including it in this work for the sake of clarity. However, to explain the main idea for the low privacy codes intuitively in a nutshell, we strategically modify the effective privacy parameter T to leverage quantum resources for rate improvement. For

interested readers, [17] considers the low privacy regime.

We focus on designing high privacy codes for different relationship between the privacy and matrix parameters, i.e., T , K and L . As a very specific example, we study the case when $K = L = T = n^2$, since this setting has a closed-form solution for the optimal number of servers in the classical setting. In the other cases, we consider several relationships, including different exponents and arithmetic differences. We compare the rates of these codes with the GASP code as it is the state-of-the-art code for classical PDMM in the high privacy regime.

Our results show that the achievable rate of our proposed quantum codes ranges from matching the classical rate to doubling it, depending on the specific relation between T , L , and K . These two extremes arise from two different reasons, the first reason is related to the optimality of the GASP codes in some asymptotic cases, e.g., as $n \rightarrow \infty$ when $T = K = L = n^2$, thus there is no room for improvement in the quantum setting due to the Holevo bound. On the other hand, the privacy being high allows the quantum codes we design to drop them upon measurements, giving a larger subspace for computing multiplications.

Owing to space limitations, this paper presents the main results together with one representative achievable scheme, while the necessary preliminaries, detailed derivations, and full proofs are deferred to [17].

II. PROBLEM FORMULATION

We now formulate the PDMM problem in the classical and quantum settings, respectively. In the classical setting, a user possessing two matrices A and B , with constrained local computational resources, seeks to compute their product. These matrices are divided into K and L sub-blocks, respectively, as follows

$$A = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_K \end{bmatrix}, \quad B = [B_1, B_2, \dots, B_L], \quad (1)$$

where all $A_i B_j$ are compatible for computing the outer product, i.e.,

$$AB = \begin{bmatrix} A_1 B_1 & A_1 B_2 & \dots & A_1 B_L \\ A_2 B_1 & A_2 B_2 & \dots & A_2 B_L \\ \vdots & \vdots & \ddots & \vdots \\ A_K B_1 & A_K B_2 & \dots & A_K B_L \end{bmatrix}. \quad (2)$$

Given K and L , the user distributes the encoded versions of the matrices to N servers, such that these matrices are secure, i.e., private, against any T colluding servers, i.e.,

$$I(f_{\mathcal{T}_1}, g_{\mathcal{T}_2}; A, B) = 0, \quad (3)$$

where f_i and g_i are the information sent by the user to the i th server about A and B , respectively.¹ In addition, $\mathcal{T}_i \subset [N]$, with $|\mathcal{T}_i| \leq T$, $i \in [2]$. It is important to note that T collusion here signifies that any T servers can share the received information from the user.

In order to ensure privacy for each of these matrices, a subspace with dimension at least T has to be designed to protect the transmitted matrices. This can potentially result in a noise space with high dimensionality. Without careful design, the resulting code may suffer rate inefficiency. To address this, we adopt polynomial codes: Polynomial codes are considered as a special case of linear codes due to their nice structure that can be utilized in the analysis.

A polynomial code for product computation is defined by two polynomials, $f : \mathbb{F}_q \rightarrow \mathbb{F}_q^{s \times \kappa}$ and $g : \mathbb{F}_q \rightarrow \mathbb{F}_q^{\kappa \times \ell}$, and the goal is to compute $h = fg$ using the N servers by sending distinct evaluations of each of these polynomials to the servers, i.e., $f_i = f(x_i)$ and $g_i = g(x_i)$ are sent to the i th server. Classically, once the user receives the responses, $h_{[N]}$, the required multiplication can be reconstructed, i.e.,

$$H(AB|h_{[N]}) = 0, \quad (4)$$

where h_i is the multiplication done at the i th server. In addition, the rate can be defined as

$$R_C(K, L, T) = \frac{KL}{N}. \quad (5)$$

Remark 1 *It is important to note that, since only one polynomial, f , is used to encode the sub-blocks of A_i , sub-blocks have to be of equal dimensions, or zero padding must be used. The same argument holds for the sub-blocks of B .*

In the quantum setting, the servers share an entangled state, represented by the density matrix ρ^{init} , in the quantum system $\mathcal{Q}^{\text{init}} = \mathcal{Q}_1^{\text{init}} \cdot \mathcal{Q}_2^{\text{init}} \dots \mathcal{Q}_N^{\text{init}}$, where the $\mathcal{Q}_n^{\text{init}}$ subsystem is at a state represented by the density matrix $\rho_n^{\text{init}} = \text{tr}_{j \in [N] \setminus \{n\}}(\rho^{\text{init}})$, where $\text{tr}_m(\cdot)$ is the partial trace operator in the finite Hilbert space. The user sends the sub-blocks' encoded versions, f_n and g_n , to the n th server over a classical noiseless channel analogous to the classical PDMM protocol. Once the n th server receives the encoded versions, it can apply classical operations and encode over its own quantum state, ρ_n^{init} , to produce another quantum state ρ_n , i.e.,

$$\rho_n = \text{Enc}(\rho_n^{\text{init}}, f_n, g_n). \quad (6)$$

Then, each server transmits ρ_n to the user over a noise-free quantum channel. Upon receiving all quantum subsystems, the user applies an appropriate measurement to recover the classical output, Y , from the quantum states to decode the required multiplication, AB , i.e.,

$$H(AB|Y) = 0. \quad (7)$$

¹It is assumed that A and B are generated independently and uniform at random.

Remark 2 Our encoding and decoding approaches follow the N -sum-box framework described in [18].

The efficiency, i.e., rate, for any scheme satisfying (3) and (7) is defined as the ratio between the number of computations and the number of downloaded symbols, i.e.,

$$R(K, L, T) = \frac{mKL}{N}, \quad (8)$$

where m is an integer representing the number of instances for AB that can be computed. This is a generalization for the rate formulation; however, in the classical setting, we have $m = 1$ and the classical rate is defined as $R_C(K, L, T)$. In the quantum setting, $m = 2$, which will become clear in later sections. The quantum rate is denoted as $R_Q(K, L, T)$. It is important to note that the number of servers in the quantum setting can potentially be different than the number of servers in the classical setting.

III. MAIN RESULTS AND DISCUSSIONS

We first note that for all the schemes, we define the vector $\alpha = (\alpha_1, \alpha_2)$, similarly for $\beta = (\beta_1, \beta_2)$.

A. High Privacy Regime Codes

Theorem 1 Let $K = L = T = n^2$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^2, n^2, n^2) = \frac{2n^4}{2n^4 + 2n^2 - 1}, \quad (9)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^2 - 1], \quad (10)$$

$$\alpha_2 = [n^4, n^4 + 1, \dots, n^4 + n^2 - 1], \quad (11)$$

$$\beta_2 = [2n^2 - 1, 3n^2 - 1, \dots, n^4 + n^2 - 1], \quad (12)$$

and the following encoding polynomials

$$f(x) = \sum_{i=1}^{n^2} R_i x^{\alpha(i)} + \sum_{i=1}^{n^2} A_i x^{\alpha(i+n^2)}, \quad (13)$$

$$g(x) = \sum_{i=1}^{n^2} Z_i x^{\beta(i)} + \sum_{i=1}^{n^2} B_i x^{\beta(i+n^2)}. \quad (14)$$

Theorem 2 Let $K = L = n^k$, and $T = n^m$, with $m \geq k \geq 2$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^k, n^m) = \frac{2n^{2k}}{2n^{2k} + 3n^m - n^k - 1}, \quad (15)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^m - 1], \quad (16)$$

$$\alpha_2 = [n^m + n^{2k} - n^k, n^m + n^{2k} - n^k + 1, \dots, n^m + n^{2k} - 1], \quad (17)$$

$$\beta_2 = [2n^m - 1, 2n^m + n^k - 1, \dots, 2n^m + n^{2k} - n^k - 1], \quad (18)$$

and the following encoding polynomials

$$f(x) = \sum_{i=1}^{n^m} R_i x^{\alpha(i)} + \sum_{i=1}^{n^k} A_i x^{\alpha(i+n^m)}, \quad (19)$$

$$g(x) = \sum_{i=1}^{n^m} Z_i x^{\beta(i)} + \sum_{i=1}^{n^k} B_i x^{\beta(i+n^m)}. \quad (20)$$

To make it easier to visualize, we add Table I for convenience. In Table I, the upper left $n^2 \times n^2$ is the information subspace and the rest is the noise subspace.

Remark 3 Although Theorem 1 is a special case of Theorem 2, it is presented separately for clarity of comparison. The former is the only closed-form solution for the number of servers in the main GASP paper [15]. Thus, we consider it a stand-alone result.

Theorem 3 Let $K = L = n^k$, and $T = n^k + r$, with $0 \leq r < n^{2k} - n^k + 1$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^k, n^k + r) = \frac{2n^{2k}}{2n^{2k} + 2n^k + 2r - 1}, \quad (21)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^k + r - 1], \quad (22)$$

$$\alpha_2 = [n^{2k}, n^{2k} + 1, \dots, n^{2k} + n^k - 1], \quad (23)$$

$$\beta_2 = [2n^k + 2r - 1, 3n^k + 2r - 1, \dots, n^{2k} + n^k + 2r - 1], \quad (24)$$

and the following encoding polynomials

$$f(x) = \sum_{i=1}^T R_i x^{\alpha(i)} + \sum_{i=1}^K A_i x^{\alpha(i+T)}, \quad (25)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{i=1}^L B_i x^{\beta(i+T)}. \quad (26)$$

Remark 4 The previous three theorems, Thms. 1, 2 and 3 analyze the case when $T \geq K \geq L$. In the first two the exponents difference is considered. The last one considers the arithmetic difference.

Theorem 4 For $K \geq L = T$, then there exists a quantum scheme based on the polynomial code that achieves a rate equal to

$$R_Q(K, L, T) = \frac{2KT}{2KT + 2T - 1}, \quad (27)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, T - 1], \quad (28)$$

$$\alpha_2 = [2T - 1, 3T - 1, \dots, KT + T - 1], \quad (29)$$

$$\beta_2 = [KT, KT + 1, \dots, KT + T - 1], \quad (30)$$

and the encoding polynomials as in (25) and (26).

$\alpha \backslash \beta$	0	1	$\dots$	$n^2 - 1$	$2n^2 - 1$	$3n^2 - 1$	$\dots$	$n^4 + n^2 - 1$
0	0	1	$\dots$	$n^2 - 1$	$2n^2 - 1$	$3n^2 - 1$	$\dots$	$n^4 + n^2 - 1$
1	1	2	$\dots$	n^2	$2n^2$	$3n^2$	$\dots$	$n^4 + n^2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^2 - 1$	$n^2 - 1$	n^2	$\dots$	$2n^2 - 2$	$3n^2 - 2$	$4n^2 - 2$	$\dots$	$n^4 + 2n^2 - 2$
n^4	n^4	$n^4 + 1$	$\dots$	$n^4 + n^2 - 1$	$n^4 + 2n^2 - 1$	$n^4 + 3n^2 - 1$	$\dots$	$2n^4 + n^2 - 1$
$n^4 + 1$	$n^4 + 1$	$n^4 + 2$	$\dots$	$n^4 + n^2$	$n^4 + 2n^2$	$n^4 + 3n^2$	$\dots$	$2n^4 + n^2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^4 + n^2 - 1$	$n^4 + n^2 - 1$	$n^4 + n^2$	$\dots$	$n^4 + 2n^2 - 2$	$n^4 + 3n^2 - 2$	$n^4 + 4n^2 - 2$	$\dots$	$2n^4 + 2n^2 - 2$

TABLE I: Degree table for $K = L = T = n^2$.

Theorem 5 Let $K = T = n^k$ and $L = n^\ell$ with $k \geq \ell$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^\ell, n^k) = \frac{2n^{k+\ell}}{2n^{k+\ell} + 2n^k - 1}, \quad (31)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^k - 1], \quad (32)$$

$$\alpha_2 = [n^{\ell+k}, n^{\ell+k} + 1, \dots, n^{\ell+k} + n^k - 1], \quad (33)$$

$$\beta_2 = [2n^k - 1, 3n^k - 1, \dots, n^{k+\ell} + n^k - 1], \quad (34)$$

and the encoding polynomials as in (25) and (26).

Corollary 1 Let $K = T = 2^3$ and $L = 2$, then

$$\frac{R_Q(8, 2, 8)}{R_C(8, 2, 8)} = 2. \quad (35)$$

Theorem 6 Let $K = T = n^\ell + r$ and $L = n^\ell$ with $r > 0$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^\ell + r, n^\ell, n^\ell + r) = \frac{2n^{2\ell} + 2rn^\ell}{2n^{2\ell} + 2rn^\ell + 2n^\ell + 2r - 1}, \quad (36)$$

with the following exponents

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^\ell + r - 1], \quad (37)$$

$$\alpha_2 = [2n^\ell + r - 1, 3n^\ell + r - 1, \dots, n^{2\ell} + (r + 1)n^\ell + r - 1], \quad (38)$$

$$\beta_2 = [n^{2\ell} + rn^\ell + r, n^{2\ell} + rn^\ell + r + 1, \dots, n^{2\ell} + (r + 1)n^\ell + r - 1], \quad (39)$$

and the encoding polynomials as in (25) and (26).

Remark 5 Theorems 4, 5 and 6 are designed for the case when $K \geq T \geq L$. In Theorem 4, the general scheme for arbitrary K , L and T is designed. For Theorems 5 and 6,

more specialized and efficient schemes are designed for the cases of exponent difference and arithmetic difference.

Remark 6 Note the difference between Theorems 2 and 5, where the first is when $K = L$, and the latter is when $K = T$. Similarly, for Theorems 3 and 6.

Remark 7 Although Theorem 6 is a general case of Theorem 5, when substituting $r = n^k - n^\ell$, the number of required servers is more in the latter compared to the former, which decreases the rate. Thus, the designed code for the special case in Theorem 5 is more optimized than the general one in Theorem 6. It is exactly the same case for Theorem 2 and Theorem 3.

Remark 8 All the rate comparisons here are with respect to the classical GASP scheme.

As mentioned earlier, Theorem 1 is the only theorem that we can compare analytically with the GASP code since it is the only closed-form result. First, let us recall the GASP code result for the same setting.

Theorem 7 (Proposition 1 in [15]) For $K = L = T = n^2$, the optimal rate for GASP when $n > 1$ is given by

$$R_C(n^2, n^2, n^2) = \frac{n^4}{n^4 + 2n^3 + 2n^2 - n - 2}. \quad (40)$$

Thus, the ratio between the classical and the quantum schemes in this case is given by

$$\frac{R_Q(n^2, n^2, n^2)}{R_C(n^2, n^2, n^2)} = \frac{2n^4 + 4n^3 + 4n^2 - 2n - 4}{2n^4 + 2n^2 - 1}. \quad (41)$$

In comparison with the classical GASP, Figs. 1-5 are plotted to compare the cases of Theorems 2 and 5. We chose both specifically since one of them has T greater than both K and L , while the other has T equal to K and greater than L . In the first case, Figs. 1 and 2 show that the efficiency of

our scheme increases monotonically with T until $T = KL$, with fixed K and L . The efficiency always goes to 1 in the case when $T = o(KL)$ since the classical GASP scheme is optimal in the asymptotic sense. However, as T increases, our scheme becomes more efficient than the GASP code even in the asymptotic sense and achieves the highest efficiency when $T = KL$. In addition, in Figs. 3 and 4, the schemes developed can be unnecessary once $m = 2k$ since the feasibility condition is then definitely satisfied, thus achieving the super-dense coding gain. Figs. 3 and 4 show that whenever L increases, the rate gain diminishes for fixed $K = T$.

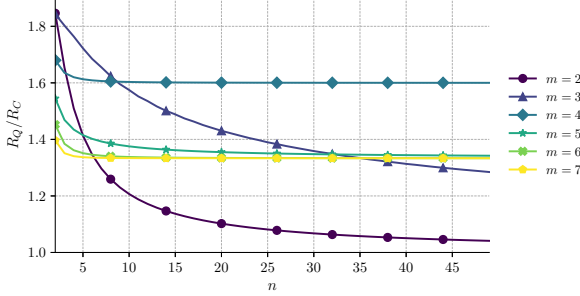


Fig. 1: $K = L = n^2$ and $T = n^m$, for $m = [2 : 7]$.

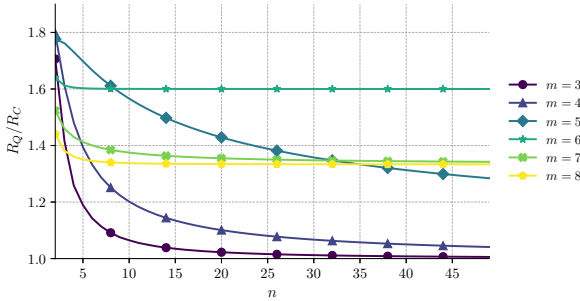


Fig. 2: $K = L = n^3$ and $T = n^m$, for $m = [3 : 8]$.

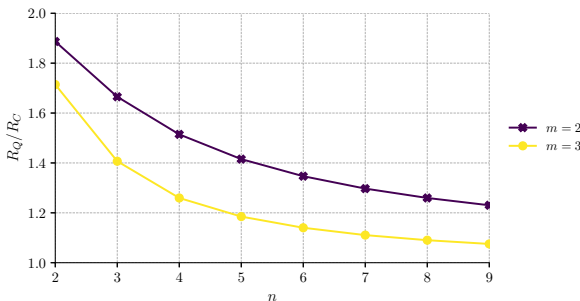


Fig. 3: $K = T = n^4$ and $L = n^m$, for $m = 2, 3$.

IV. SCHEME EXAMPLE

As an illustrative example, we provide a detailed derivation of the transmitted and received symbols along with the chosen

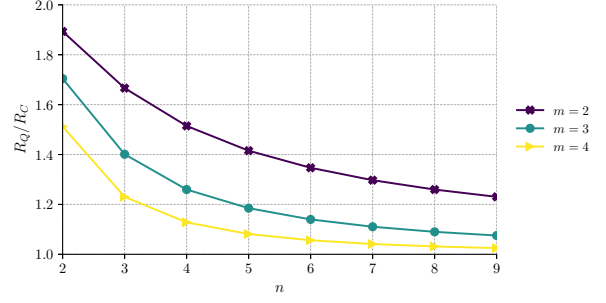


Fig. 4: $K = T = n^5$ and $L = n^m$, for $m = [2 : 4]$.

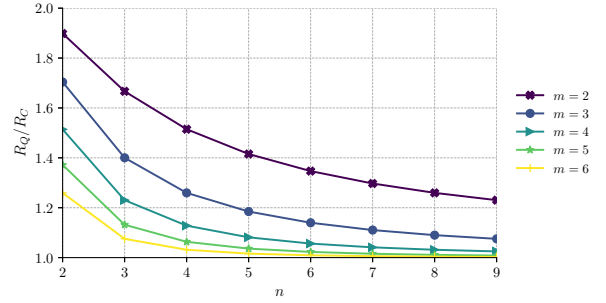


Fig. 5: $K = T = n^7$ and $L = n^m$, for $m = [2 : 6]$.

transfer matrix for the case when $K = L = T = n^2$ in Theorem 1. The user sends the two polynomials $f(x)$ and $g(x)$ defined as

$$f(a_i) = \sum_{k=1}^K A_k a_i^{\alpha_k} + \sum_{t=1}^T R_t a_i^{\alpha_{K+t}}, \quad (42)$$

$$g(a_i) = \sum_{\ell=1}^L B_\ell a_i^{\beta_\ell} + \sum_{t=1}^T S_t a_i^{\beta_{L+t}}, \quad (43)$$

to each of the N servers. Upon receiving them, each server applies the multiplication and computes

$$F^{[m]}(a_n) = \sum_{(k,\ell) \in \mathcal{Q}} A_k^{[m]} B_\ell^{[m]} a_n^{(\alpha \oplus \beta)_{k,\ell}} + \sum_{(i,j) \in \mathcal{Q}} I_{i,j}^{[m]} a_n^{(\alpha \oplus \beta)_{i,j}}, \quad m \in [2], \quad (44)$$

where $\mathcal{Q}$ is the interference subspace with $i \in [K]$, or $j \in [L]$. Now, for each element in $F_n^{[1]} = F^{[1]}(a_n)$, $F_n^{[2]} = F^{[2]}(a_n)$, the server applies Pauli operators on each element as follows

$$\begin{aligned} & X(u_n F_n^{[1]}(i,j)) Z(v_n F_n^{[2]}(i,j)) \rho_n \\ & \times X^\dagger(u_n F_n^{[1]}(i,j)) Z^\dagger(v_n F_n^{[2]}(i,j)). \end{aligned} \quad (45)$$

The transmitted symbols can be written as follows,

$$\begin{aligned} X(i,j) = & [F_1^{[1]}(i,j), \dots, F_{2n^4+2n^2-1}^{[1]}(i,j), \\ & F_1^{[2]}(i,j), \dots, F_{2n^4+2n^2-1}^{[2]}(i,j)]^t \end{aligned} \quad (46)$$

$$\begin{aligned}
&= \begin{bmatrix} \text{diag}(u)G' & 0_{2n^4+2n^2-1} \\ 0_{2n^4+2n^2-1} & \text{diag}(v)G' \end{bmatrix} \\
&\times \left[\underbrace{I_1^{[1]}(i,j), \dots, I_{n^4+2n^2-1}^{[1]}(i,j), (A_1 B_1)^{[1]}(i,j),}_{\text{Interference space 1}} \right. \\
&\quad (A_1 B_2)^{[1]}(i,j), \dots, (A_K B_L)^{[1]}(i,j), \\
&\quad \underbrace{I_1^{[2]}(i,j), \dots, I_{n^4+2n^2-1}^{[2]}(i,j), (A_1 B_1)^{[2]}(i,j),}_{\text{Interference space 2}} \\
&\quad \left. \dots, (A_K B_L)^{[2]}(i,j) \right]^t, \tag{47}
\end{aligned}$$

where

$$G' = \begin{bmatrix} 1 & a_1 & a_1^2 & \dots & a_1^{2n^4+2n^2-2} \\ 1 & a_2 & a_2^2 & \dots & a_2^{2n^4+2n^2-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_{2n^4+2n^2-1} & a_{2n^4+2n^2-1}^2 & \dots & a_{2n^4+2n^2-1}^{2n^4+2n^2-2} \end{bmatrix}. \tag{48}$$

There are two interference vector spaces as shown in (46), each for a separate classical instance of the code, i.e., two multiplications, $AB^{[1]}$ and $AB^{[2]}$. The transfer matrix M for the N -sum box should be chosen such that the maximum number of symbols in the first and second interference vector spaces are dropped. This must be done while using a valid transfer matrix M , i.e., full rank and satisfying the SSO condition. Upon designing the appropriate transfer matrix, M , the symbols that the user gets after measurement can be written as,

$$\begin{aligned}
Y(i,j) &= MX(i,j) \\
&= \left[\underbrace{I_{\lfloor \frac{2n^4+2n^2-1}{2} \rfloor}^{[1]}(i,j), \dots, I_{n^4+2n^2-1}^{[1]}(i,j),}_{\text{Remaining interference subspace 1}} \right. \\
&\quad (A_1 B_1)^{[1]}(i,j), \dots, (A_{n^2} B_{n^2})^{[1]}(i,j), \\
&\quad \underbrace{I_{\lceil \frac{2n^4+2n^2-1}{2} \rceil}^{[2]}(i,j), \dots, I_{n^4+2n^2-1}^{[2]}(i,j),}_{\text{Remaining interference subspace 2}} \\
&\quad \left. (A_1 B_1)^{[2]}(i,j), \dots, (A_{n^2} B_{n^2})^{[2]}(i,j) \right]^t, \tag{49}
\end{aligned}$$

where $M = \begin{bmatrix} 0 & I_N \end{bmatrix} [G \ H]^{-1}$, with

$$G = D \begin{bmatrix} G'_1 & 0 \\ 0 & G'_2 \end{bmatrix}, \tag{51}$$

with $G'_1 = G'(:, 1 : \lfloor \frac{2n^4+2n^2-1}{2} \rfloor - 1)$, $G'_2 = G'(:, 1 : \lceil \frac{2n^4+2n^2-1}{2} \rceil - 1)$ and

$$H = D \begin{bmatrix} H_1 & 0 \\ 0 & H_2 \end{bmatrix}, \tag{52}$$

with $H_1 = G'(:, \lfloor \frac{2n^4+2n^2-1}{2} \rfloor : \text{end})$, $H_2 = G'(:, \lceil \frac{2n^4+2n^2-1}{2} \rceil : \text{end})$, and

$$D = \begin{bmatrix} \text{diag}(u) & 0 \\ 0 & \text{diag}(v) \end{bmatrix}. \tag{53}$$

As shown in (49), the remaining interference subspace has lower dimensionality compared to the one in (46). The first interference subspace in (49) starts from $\lfloor \frac{2n^4+2n^2-1}{2} \rfloor$ as opposed from starting at 1 in (46). Similarly, for the second interference subspace in (49) starts from $\lceil \frac{2n^4+2n^2-1}{2} \rceil$. Thus, from (49), upon measurement, the user receives two instances of AB , i.e., $AB^{[1]}$, and $AB^{[2]}$, over a subspace of dimension $2n^4$.

Remark 9 The remaining achievable schemes for Theorems 2 -6 follow the same main steps as the one represented in this section for $T = K = L = n^2$, with the difference of the chosen exponents.

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